BOSTON UNIVERSITY

MET AD 571 Assignment #2

Professor Gregory Page

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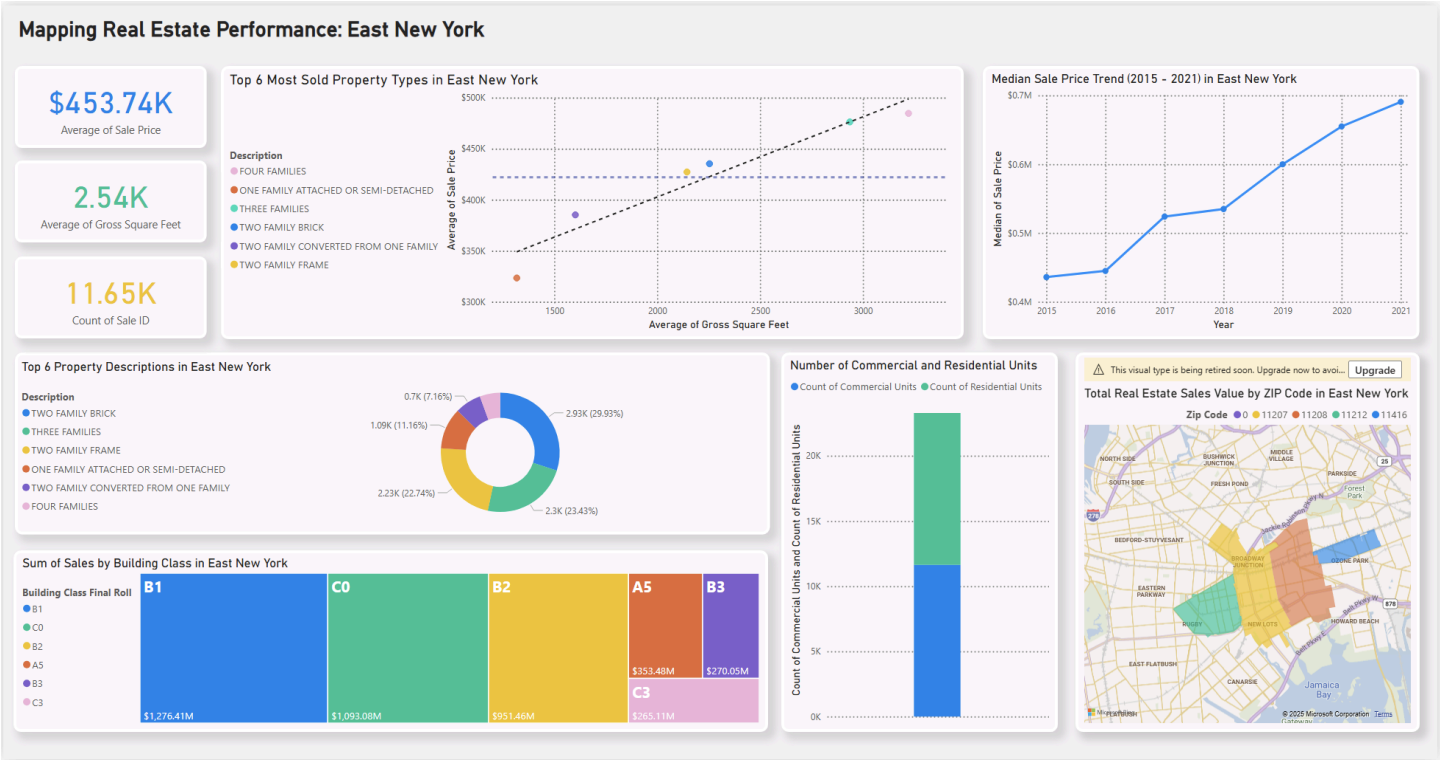
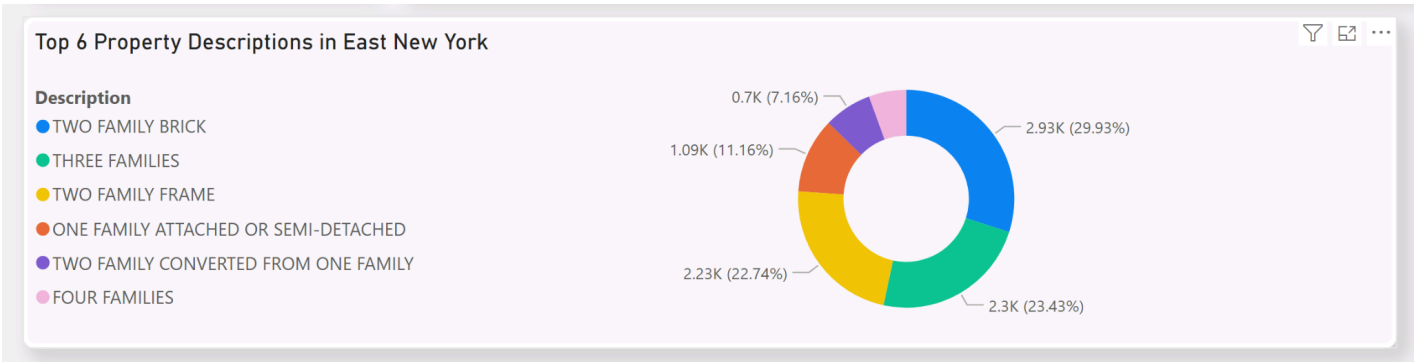
Dashboard Overview



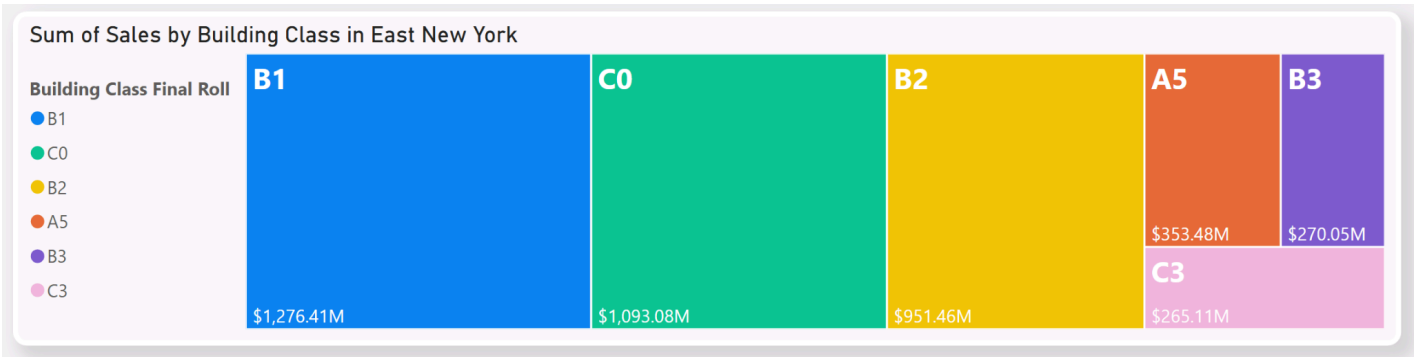
This Power BI dashboard provides a comprehensive overview of the real estate market in East New York, using six distinct visualizations to highlight key patterns and trends. The visuals collectively explore property composition, sales distribution, pricing behavior, and geographic differences. The donut and tree map charts identify the most common and highest-value property types, while the stacked bar chart shows the dominance of residential units over commercial ones. The filled map highlights sales activity across ZIP codes, revealing geographic variations in total sales value. The line chart tracks the steady rise in median sale prices over time, illustrating long-term market appreciation, and the scatter plot explores how property size relates to sale price across the top-selling property types. Finally, the summary cards provide quick insights into overall market metrics. Together, these visuals offer a clear, data-driven understanding of East New York's housing market, showing how property characteristics, location, and time influence market performance and investment potential.

Pie Chart



This donut chart shows the top six property descriptions in East New York, highlighting the most common types of residential buildings. It was created in Power BI by using a donut chart visual with Description as the category and count of properties as the value, filtered using the Top N filter to show only the top six categories. Two Family Brick homes make up approximately 30% of all sales, followed by Three Families at 23% and Two Family Frame at 23%. Together, these top three categories account for nearly 80% of all property sales, confirming that multi-family homes dominate the East New York housing market. This visualization helps quickly identify key property trends and understand the overall housing composition in East New York.

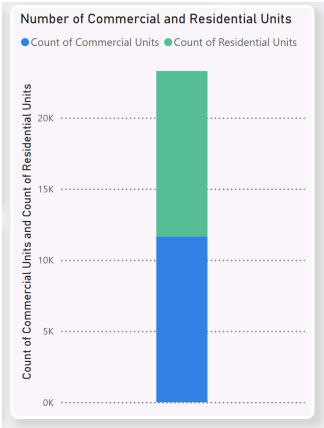
Tree Map Visualization



This treemap shows the sum of property sales by building class in East New York, providing a deeper look into the market value behind the property types shown in the first chart. Each building class represents a code corresponding to specific property descriptions (such as Two Family Brick or Three Families), allowing for a more detailed view of how different property types contribute to total sales. To create this visualization, the Building Class Final Roll field was used as the category, Sum of Sale Price as the value, and a Top N filter was applied to display the building classes with the highest total sales. Building Class B1 leads with a total sales value of about $1.2 billion, followed

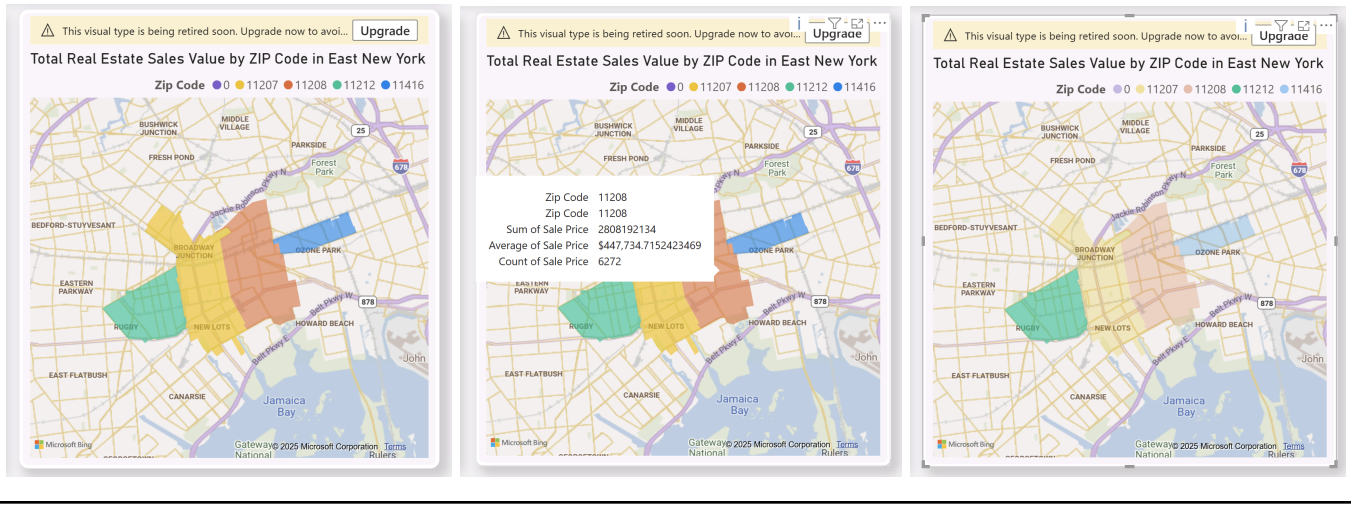
by C0 and B2, which together make up an additional $1.8 billion. These top classes collectively represent over 60% of East New York's total market value, showing how a few property types drive most of the area's real estate economy. Together with the first chart, this provides a clearer understanding of both the frequency and total market value of various property types, showing which kinds of buildings sell the most and have the greatest financial impact in the area.

Bar Chart Visualization



This stacked bar chart compares the number of residential and commercial units in East New York, showing a clear imbalance between the two. The visualization was created by counting the total number of units for each type, with residential and commercial categories stacked to show their combined total. The chart reveals that East New York contains approximately 12,000 commercial units and about 24,000 residential units, meaning there are roughly twice as many residential units as commercial ones. This confirms that the neighborhood is overwhelmingly residential, with minimal commercial infrastructure. This finding aligns with the previous charts, where residential building classes and property types (like Two Family Brick and Three Families) dominated both sales volume and total market value. Together, these visuals reinforce that East New York's real estate market is largely driven by residential properties, emphasizing their greater market share and economic influence compared to commercial assets.

Filled Map Chart Visualization



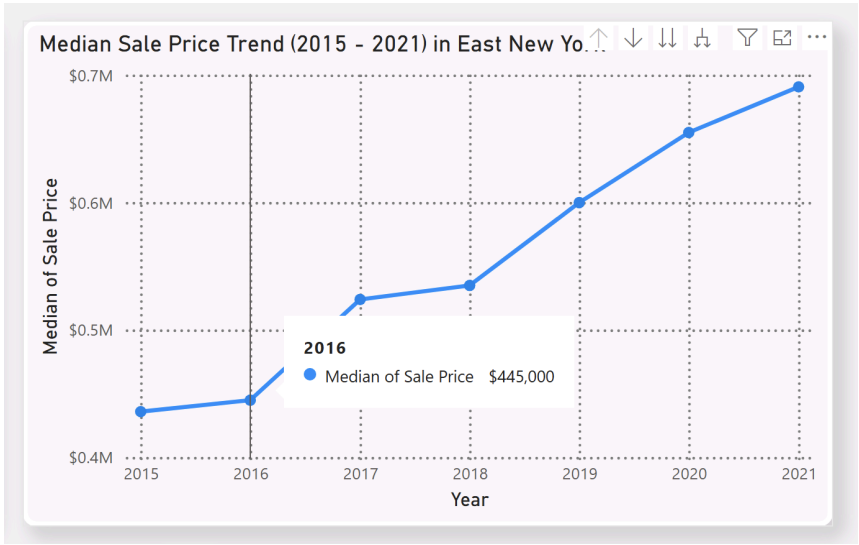
This map visualization displays the Total Real Estate Sales Value by ZIP Code within the East New York area. Each ZIP Code region (11207, 11208, 11212, and 11416) is shaded in a different color, allowing for an easy geographic comparison of real estate activity. The shading represents the total sales value aggregated by ZIP Code, showing which areas have higher or lower sales totals. For instance, ZIP Code 11208, highlighted in orange, shows a high total sales value of approximately $2.8 billion, with over 6,200 sales and an average sale price of around $447,000. This suggests strong property transaction activity in that ZIP Code compared to surrounding areas.

This visualization shows total real estate sales values by ZIP Code within East New York. Each colored region represents a ZIP Code boundary, and when hovered over, the tooltip displays additional metrics like the total sum of sale prices, the average sale price, and the count of transactions for that ZIP Code.

The visualization was created in Power BI using a Filled Map visual. The dataset includes columns for ZIP_CODE, and SALE_PRICE. The ZIP Code field was categorized as Postal Code under the Data Category settings to ensure geographic recognition. The Sum, Average, and Count of Sale Price was used as tooltips, and ZIP Code was placed in the Location field, allowing Power BI to shade each ZIP region according to total sales value.

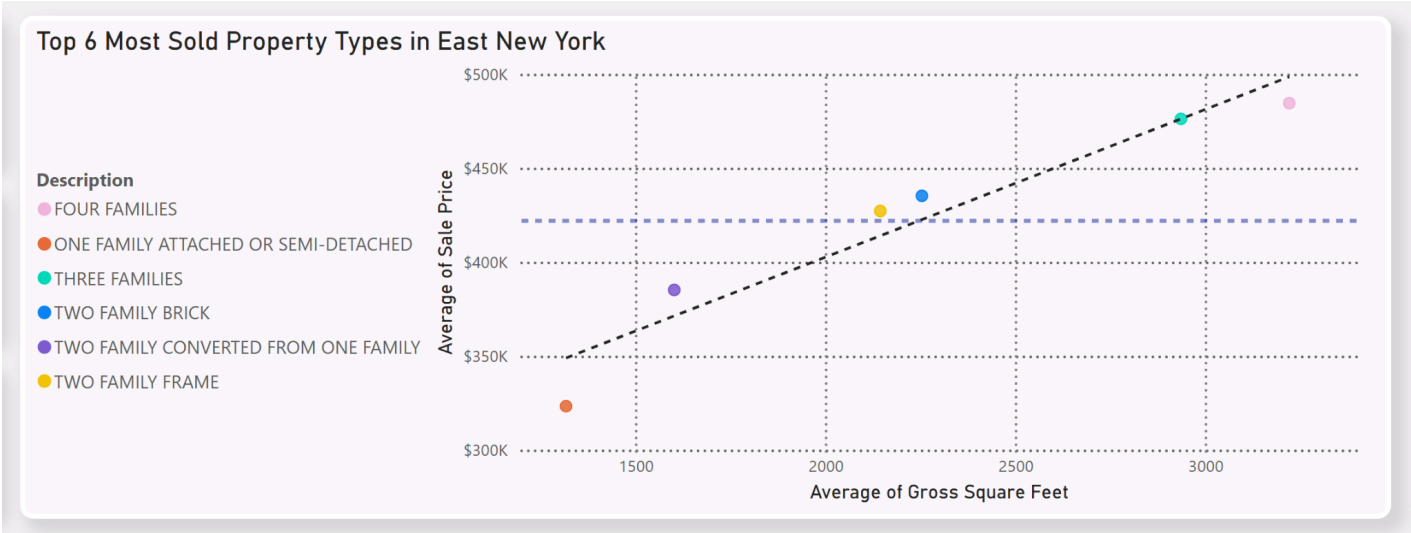
This map provides a geographic perspective on real estate trends, revealing spatial patterns that are not visible in tables or bar charts. It helps decision-makers and analysts quickly pinpoint hotspots of real estate activity and compare neighborhood performance within the East New York area.

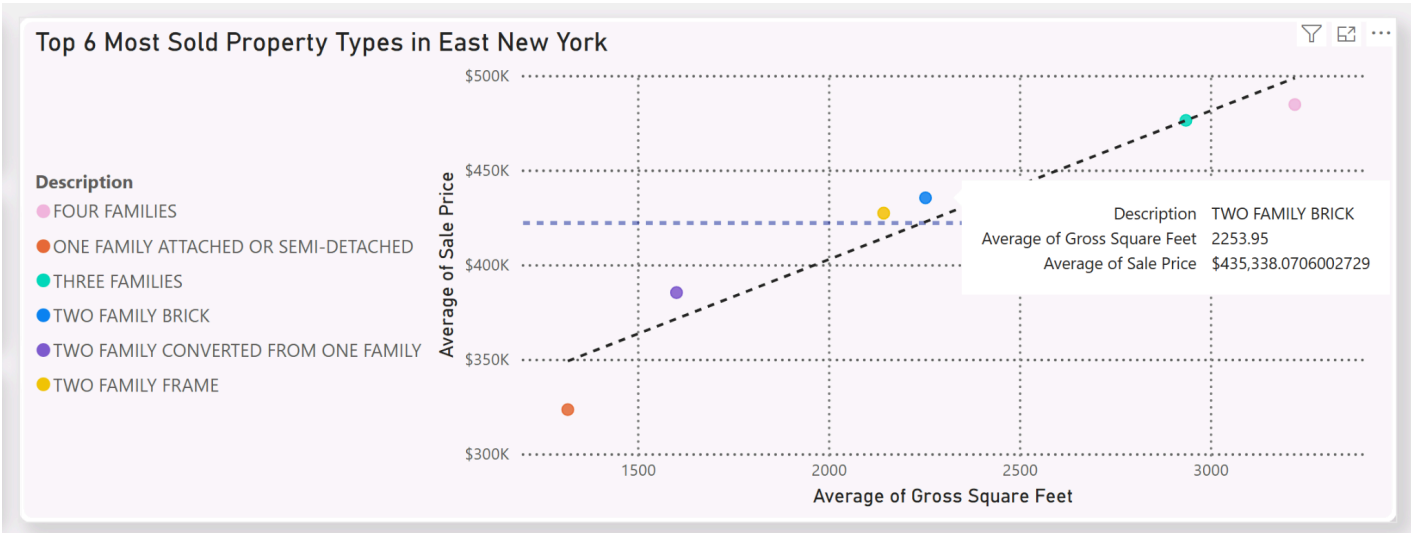
Line Chart Visualization



This line chart visualization shows the median sale price trend in East New York from 2015 to 2021, highlighting how property values have changed over time. It was created in Power BI using a Line Chart visual, with Sale Date narrowed down to Year only on the X-axis and Median of Sale Price on the Y-axis. The median was chosen instead of the average because it provides a more accurate representation of central pricing trends, reducing the influence of extreme outliers or unusually high-value sales that could skew the data. The median sale price increased from about $400,000 in 2015 to nearly $700,000 in 2021, representing a growth of approximately 75% over six years, an average increase of about $50,000 per year. This consistent growth indicates strong market appreciation in East New York, likely driven by rising demand, urban development, and broader economic factors. Overall, the chart provides a clear view of how housing prices have evolved over time, helping identify long-term trends and the overall direction of the neighborhood's real estate market.

Scatter Plot Visualization

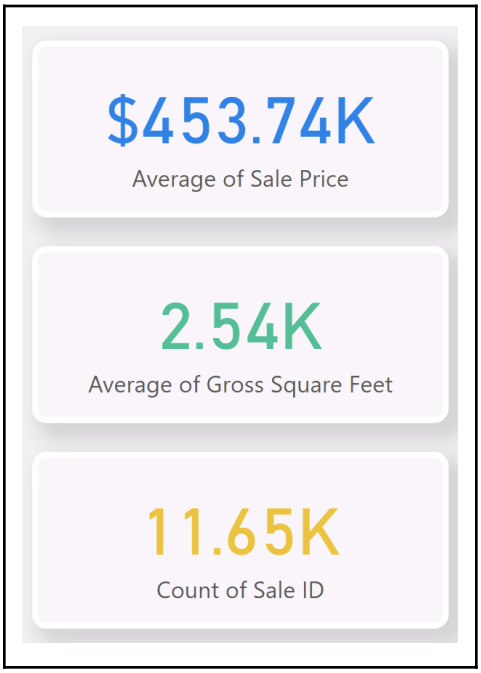




This scatter plot visualization compares the six most frequently sold property types in East New York, illustrating how the average gross square feet (X-axis) relates to the average sale price (Y-axis). Created using a Scatter Chart visual, the data was filtered to include only the top six property types by sales id count, with each type represented by a distinct color in the legend. Displaying average values helps highlight overall pricing and size trends while minimizing the impact of outliers. The trend line indicates a positive relationship between property size and sale price, larger properties generally command higher prices. For instance, “Four Families” and “Three Families” properties tend to have both greater square footage and higher average sale prices, while “One Family Attached or Semi-Detached” homes are smaller and sell for less. Notably, “Two Family Brick” homes achieve relatively high prices despite their moderate size, suggesting strong demand or construction quality. These insights demonstrate that property type influences pricing beyond just

square footage, helping identify which segments offer better value or return on investment. Overall, this visualization clearly shows how property size and type interact to shape market trends, deepening the understanding of price differences and market behavior across East New York's property landscape.

Cards Visualization



This card visualization enhances the dashboard by providing a quick snapshot of the key metrics before exploring more detailed visuals. Built using Power BI's Card visual, these summary indicators give users an instant understanding of overall market trends in East New York. By presenting the most important figures upfront, the cards set the context for the rest of the dashboard, helping viewers interpret the deeper insights and patterns shown in the subsequent charts and maps.